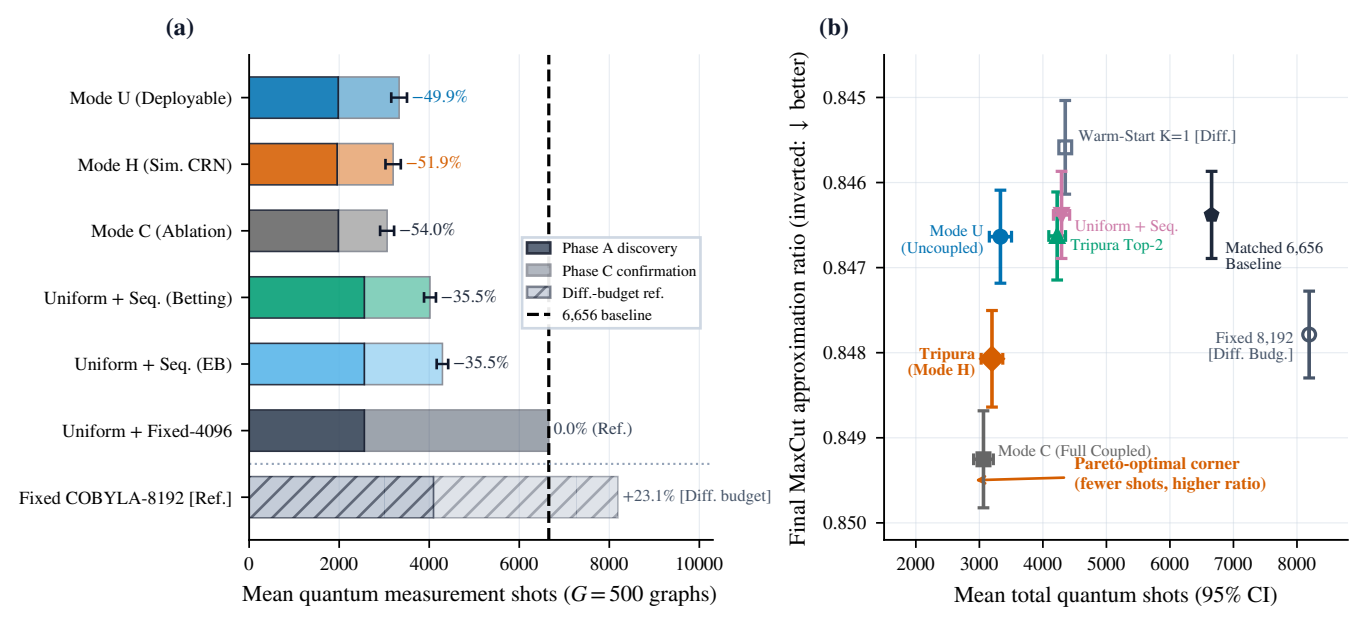
(a)
Mode U (Deployable)
Mode H (Sim. CRN)
Mode C (Ablation)
Uniform + Seq. (Betting)
Uniform + Seq. (EB)
Uniform + Fixed-4096
Fixed COBYLA-8192 [Ref.]
−49.9%
−51.9%
−54.0%
−35.5%
−35.5%
0.0% (Ref.)
+23.1% [Diff. budget]
Phase A discovery
Phase C confirmation
Diff.-budget ref.
6,656 baseline
0
2000
4000
6000
8000
10000
Mean quantum measurement shots ($G = 500$ graphs)
(b)
Final MaxCut approximation ratio (inverted: ↓ better)
0.845
0.846
0.847
0.848
0.849
0.850
2000
3000
4000
5000
6000
7000
8000
Mean total quantum shots (95% CI)
Warm-Start K=1 [Diff.]
Mode U (Uncoupled)
Uniform + Seq. Tripura Top-2
Matched 6,656 Baseline
Fixed 8,192 [Diff. Budg.]
Tripura (Mode H)
Mode C (Full Coupled)
Pareto-optimal corner (fewer shots, higher ratio)